

Mathematical Methods for Finance

Module 1 – Exam-Style Exercise Pack

Prof. Daniele Angelini

University of Perugia

Academic Year 2026/2027

100 different exercises with detailed step-by-step solutions

How to use this exercise pack

Exercise: Training strategy

- ① Try the exercise before reading the solution.
- ② Write the relevant definition before using formulas.
- ③ Check the hypotheses: completeness, compactness, closedness, boundedness, linearity.
- ④ In counterexamples, identify exactly which axiom or implication fails.
- ⑤ Use the stars as a guide: ★ basic, ★★★★★ exam challenge.

Hint: Coverage

The 100 exercises cover metric spaces, completeness, compactness, normed spaces, sequence spaces, function spaces, Banach spaces, bounded linear operators, inner products, Hilbert spaces and projections.

Exercise map

- 1–18 Metric spaces and topology.
- 19–35 Completeness, contractions and compactness.
- 36–60 Normed spaces, sequence spaces and inequalities.
- 61–80 Function spaces and linear operators.
- 81–100 Inner product and Hilbert spaces.

Module 1 Exercise Pack

Metric spaces and topology

Exercises 1–18

Exercise 001 – Metric axioms on the line

Exercise: ★

On $\mathbb{R}$ define $d(x, y) = |x - y|$. Verify directly that d is a metric.

Solution 001 – Metric axioms on the line

Solution:

- ① Non-negativity is clear because absolute values are nonnegative.
 - ② $d(x, y) = 0$ means $|x - y| = 0$, hence $x = y$; conversely, if $x = y$, then $d(x, y) = 0$.
 - ③ Symmetry: $|x - y| = |y - x|$.
 - ④ Triangle inequality follows from $|x - z| = |(x - y) + (y - z)| \leq |x - y| + |y - z|$.
- Therefore d is a metric.

Exercise 002 – Squared distance

Exercise: ★★

On $\mathbb{R}$ define $d(x, y) = (x - y)^2$. Which metric axiom fails? Give explicit numbers.

Solution 002 – Squared distance

Solution:

Non-negativity, identity and symmetry hold. The triangle inequality fails. Choose $x = 0$, $y = 1$, $z = 2$. Then $d(0, 2) = 4$, while $d(0, 1) + d(1, 2) = 1 + 1 = 2$. The inequality $4 \leq 2$ is false. Hence d is not a metric.

Exercise 003 – A transformed metric

Exercise: ★ ★ ★

Let $d(x, y) = |x^3 - y^3|$ on $\mathbb{R}$. Is this a metric?

Solution 003 – A transformed metric

Solution:

Let $\varphi(x) = x^3$. Since φ is injective, $|x^3 - y^3| = 0$ iff $x = y$. Symmetry and non-negativity are immediate. For the triangle inequality, use the usual metric after transformation: $|x^3 - z^3| \leq |x^3 - y^3| + |y^3 - z^3|$. Therefore d is a metric.

Exercise 004 – A formula that forgets signs

Exercise: ★★

On $\mathbb{R}$ define $d(x, y) = |x^2 - y^2|$. Is it a metric?

Solution 004 – A formula that forgets signs

Solution:

It is nonnegative and symmetric. However, identity fails: $d(1, -1) = |1 - 1| = 0$, but $1 \neq -1$. Since different points can have zero distance, d is not a metric.

Exercise 005 – Discrete metric convergence

Exercise: ★★

Let X be any set with the discrete metric. Characterize all convergent sequences in X .

Solution 005 – Discrete metric convergence

Solution:

In the discrete metric, $d(x_n, x) < 1/2$ implies $x_n = x$. If $x_n \rightarrow x$, choose $\varepsilon = 1/2$. Then there exists N such that $n \geq N$ implies $d(x_n, x) < 1/2$, hence $x_n = x$. Thus a sequence converges iff it is eventually constant.

Exercise 006 – Bounded metric

Exercise: ★ ★ ★

Let (X, d) be a metric space and set

$$\rho(x, y) = \frac{d(x, y)}{(1 + d(x, y))}.$$

Prove that ρ is bounded and explain why it is a metric.

Solution 006 – Bounded metric

Solution:

Boundedness: for $t \geq 0$, $t/(1+t) < 1$, so $0 \leq \rho(x, y) < 1$. Metric property follows from the fact that $\varphi(t) = t/(1+t)$ is increasing, $\varphi(0) = 0$, and subadditive: $\varphi(a+b) \leq \varphi(a) + \varphi(b)$. Hence $\rho = \varphi \circ d$ is a metric.

Exercise 007 – Product metric

Exercise: ★★

Let (X, d_X) and (Y, d_Y) be metric spaces. Show that

$$D((x, y), (x', y')) = \max\{d_X(x, x'), d_Y(y, y')\}$$

is a metric on $X \times Y$.

Solution 007 – Product metric

Solution:

Non-negativity and symmetry are inherited from d_X, d_Y . If $D = 0$, both component distances are zero, so $x = x'$ and $y = y'$. For the triangle inequality, each component satisfies it: $d_X(x, z) \leq d_X(x, u) + d_X(u, z)$ and similarly for Y . Taking maxima gives $D(p, r) \leq D(p, q) + D(q, r)$.

Exercise 008 – Open balls in two metrics

Exercise: ★★

Describe $B(2, 1/2)$ in $\mathbb{R}$ with the usual metric and with the discrete metric.

Solution 008 – Open balls in two metrics

Solution:

Usual metric: $B(2, 1/2) = \{x : |x - 2| < 1/2\} = (3/2, 5/2)$. Discrete metric: since $1/2 < 1$, the only point with distance $< 1/2$ from 2 is 2 itself. Hence $B(2, 1/2) = \{2\}$.

Exercise 009 – Open sets in a subspace

Exercise: ★★

In the subspace $X = [0, 2]$ of $\mathbb{R}$, is $[0, 1)$ open in X ? Is it open in $\mathbb{R}$?

Solution 009 – Open sets in a subspace

Solution:

In X , $[0, 1) = [0, 2] \cap (-1, 1)$, so it is open in the subspace topology. In $\mathbb{R}$, it is not open because $0 \in [0, 1)$ but every usual ball around 0 contains negative points not in $[0, 1)$.

Exercise 010 – Interior, closure, boundary

Exercise: ★★

For $A = (0, 1] \cup \{2\} \subset \mathbb{R}$, compute A° , $\overline{A}$ and ∂A .

Solution 010 – Interior, closure, boundary

Solution:

The interior is $A^\circ = (0, 1)$. The closure adds the missing limit point 0 and keeps the isolated point 2: $\bar{A} = [0, 1] \cup \{2\}$. The boundary is $\partial A = \bar{A} \setminus A^\circ = \{0, 1, 2\}$.

Exercise 011 – Limit and isolated points

Exercise: ★★

Let $A = \{1/n : n \in \mathbb{N}\} \cup \{3\} \subset \mathbb{R}$. Find all limit points and isolated points.

Solution 011 – Limit and isolated points

Solution:

The sequence $1/n$ accumulates only at 0, so 0 is the only limit point. The point 3 is isolated: take radius less than $\min\{1, 3 - 1\}/2$. Each $1/n$ is isolated because neighbouring elements are separated by a positive distance. Thus isolated points are all points of A .

Exercise 012 – Convergence in $\mathbb{R}^2$

Exercise: ★

Let $x_n = (1/n, (-1)^n/n) \in \mathbb{R}^2$. Prove that $x_n \rightarrow (0, 0)$ under the Euclidean metric.

Solution 012 – Convergence in $\mathbb{R}^2$

Solution:

Compute the distance: $\|x_n\|_2 = \sqrt{(1/n)^2 + ((-1)^n/n)^2} = \sqrt{2}/n$. Since $\sqrt{2}/n \rightarrow 0$, the distance from x_n to $(0,0)$ tends to zero. Hence $x_n \rightarrow (0,0)$.

Exercise 013 – Nonconvergence with oscillation

Exercise: ★★

Does $y_n = (1/n, (-1)^n)$ converge in $\mathbb{R}^2$ with the Euclidean metric?

Solution 013 – Nonconvergence with oscillation

Solution:

No. If it converged, the second coordinate would also converge. But $(-1)^n$ has two subsequential limits, 1 and -1 . Therefore the sequence cannot converge in $\mathbb{R}^2$.

Exercise 014 – Sequential closedness

Exercise: ★★ ★

Use sequences to show that $F = [0, 1]$ is closed in $\mathbb{R}$.

Solution 014 – Sequential closedness

Solution:

Let (x_n) be any sequence in F with $x_n \rightarrow x$ in $\mathbb{R}$. Since $0 \leq x_n \leq 1$ for every n , taking limits gives $0 \leq x \leq 1$. Thus $x \in F$. By the sequential characterization of closed sets in metric spaces, F is closed.

Exercise 015 – Continuity by sequences

Exercise: ★★

Prove that $f(x) = x^2$ is continuous at $x = 3$ using sequences.

Solution 015 – Continuity by sequences

Solution:

Let $x_n \rightarrow 3$. Then $x_n^2 - 9 = (x_n - 3)(x_n + 3)$. Since $x_n \rightarrow 3$, the sequence $(x_n + 3)$ is bounded, and $(x_n - 3) \rightarrow 0$. The product tends to 0. Hence $f(x_n) \rightarrow f(3)$, so f is continuous at 3.

Exercise 016 – Uniform continuity fails

Exercise: ★ ★ ★

Show that $f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$.

Solution 016 – Uniform continuity fails

Solution:

Take $x_n = n$ and $y_n = n + 1/n$. Then $|x_n - y_n| = 1/n \rightarrow 0$. But $|f(y_n) - f(x_n)| = (n + 1/n)^2 - n^2 = 2 + 1/n^2 \rightarrow 2$. Uniform continuity would force this difference to go to 0. Contradiction.

Exercise 017 – Contraction on an interval

Exercise: ★★

On $[0, 1]$, let $T(x) = (x + 2)/4$. Show that T is a contraction and find its fixed point.

Solution 017 – Contraction on an interval

Solution:

For $x, y \in [0, 1]$, $|T(x) - T(y)| = |x - y|/4$, so the Lipschitz constant is $1/4 < 1$. A fixed point satisfies $x = (x + 2)/4$, hence $4x = x + 2$, so $x = 2/3$.

Exercise 018 – Completeness of subsets

Exercise: ★★

Which of the following are complete subsets of $\mathbb{R}$ with the usual metric: $(0, 1)$, $[0, 1]$, $\{0\} \cup \{1/n : n \in \mathbb{N}\}$, $\mathbb{R} \setminus \{0\}$?

Solution 018 – Completeness of subsets

Solution:

In $\mathbb{R}$, a subset is complete iff it is closed. $(0, 1)$ is not closed, so not complete. $[0, 1]$ is closed, so complete. $\{0\} \cup \{1/n\}$ contains its only limit point 0, so closed and complete. $\mathbb{R} \setminus \{0\}$ is not closed, so not complete.

Module 1 Exercise Pack

Completeness, contractions and compactness

Exercises 19–35

Exercise 019 – Rational incompleteness

Exercise: ★ ★ ★

Construct a Cauchy sequence in $\mathbb{Q}$ that does not converge in $\mathbb{Q}$.

Solution 019 – Rational incompleteness

Solution:

Take rational decimal truncations of $\sqrt{2}$: $1, 1.4, 1.41, 1.414, \dots$. This sequence converges to $\sqrt{2}$ in $\mathbb{R}$, hence is Cauchy in $\mathbb{R}$ and also in $\mathbb{Q}$. But $\sqrt{2} \notin \mathbb{Q}$, so the sequence has no limit in $\mathbb{Q}$. Thus $\mathbb{Q}$ is not complete.

Exercise 020 – Heine-Borel check I

Exercise: ★★

Decide whether $[0, 1] \cup \{2\}$ is compact in $\mathbb{R}$.

Solution 020 – Heine-Borel check I

Solution:

The set is bounded because it is contained in $[0, 2]$. It is closed because it is the union of two closed sets: $[0, 1]$ and $\{2\}$. By Heine-Borel in $\mathbb{R}$, closed and bounded implies compact. Hence the set is compact.

Exercise 021 – Heine-Borel check II

Exercise: ★★

Decide whether $(0, 1]$ is compact in $\mathbb{R}$.

Solution 021 – Heine-Borel check II

Solution:

The set is bounded, but it is not closed because 0 is a limit point and $0 \notin (0, 1]$. In $\mathbb{R}$, compact sets must be closed and bounded. Therefore $(0, 1]$ is not compact.

Exercise 022 – Open cover without finite subcover

Exercise: ★ ★ ★

Show directly that $(0, 1)$ is not compact using an open cover.

Solution 022 – Open cover without finite subcover

Solution:

Use the open cover $U_n = (1/n, 1)$ for $n \geq 2$. If $x \in (0, 1)$, choose $n > 1/x$; then $1/n < x$, hence $x \in U_n$. Any finite subcover uses only indices up to some N , and therefore misses every $x \in (0, 1/N]$. Thus no finite subcover exists.

Exercise 023 – Sequential compactness

Exercise: ★ ★ ★

Explain why every sequence in $[0, 1]$ has a convergent subsequence with limit in $[0, 1]$.

Solution 023 – Sequential compactness

Solution:

By Bolzano-Weierstrass, every bounded sequence in $\mathbb{R}$ has a convergent subsequence. Since all terms lie in $[0, 1]$, the sequence is bounded. Let the subsequence converge to $x \in \mathbb{R}$. Because $[0, 1]$ is closed, the limit belongs to $[0, 1]$.

Exercise 024 – Compact image

Exercise: ★★

Let K be compact and $f : K \rightarrow \mathbb{R}$ continuous. Prove that $f(K)$ is compact.

Solution 024 – Compact image

Solution:

Take any sequence (y_n) in $f(K)$. Write $y_n = f(x_n)$ with $x_n \in K$. Since K is compact, a subsequence x_{n_j} converges to some $x \in K$. By continuity, $y_{n_j} = f(x_{n_j}) \rightarrow f(x) \in f(K)$. Thus every sequence in $f(K)$ has a convergent subsequence in $f(K)$, so $f(K)$ is compact.

Exercise 025 – Extreme value theorem

Exercise: ★★

Use compactness to justify that $f(x) = x(1 - x)$ attains a maximum on $[0, 1]$, then find it.

Solution 025 – Extreme value theorem

Solution:

Since $[0, 1]$ is compact and f is continuous, f attains a maximum. Compute $f(x) = x - x^2$, so $f'(x) = 1 - 2x$. The critical point is $x = 1/2$. Values: $f(0) = 0$, $f(1) = 0$, $f(1/2) = 1/4$. Maximum is $1/4$.

Exercise 026 – Infinite discrete space

Exercise: ★★☆☆

Let X be infinite with the discrete metric. Show that X is not compact.

Solution 026 – Infinite discrete space

Solution:

In the discrete metric every singleton $\{x\}$ is open. The family $\{\{x\} : x \in X\}$ is an open cover of X . A finite subcover would cover only finitely many points. Since X is infinite, no finite subcover exists. Thus X is not compact.

Exercise 027 – Closed subset of compact set

Exercise: ★ ★ ★

Prove that a closed subset F of a compact metric space K is compact.

Solution 027 – Closed subset of compact set

Solution:

Let (x_n) be a sequence in F . Since K is compact, there is a subsequence $x_{n_j} \rightarrow x \in K$. Since F is closed and all $x_{n_j} \in F$, the limit satisfies $x \in F$. Hence every sequence in F has a subsequence converging in F , so F is compact.

Exercise 028 – Contraction theorem iteration

Exercise: ★ ★ ★

Let $T(x) = \cos(x)/2$ on $[0, 1]$. Show that the Banach fixed point theorem applies.

Solution 028 – Contraction theorem iteration

Solution:

First $T([0, 1]) \subset [\cos(1)/2, 1/2] \subset [0, 1]$. Also $|T'(x)| = |-\sin(x)/2| \leq 1/2 < 1$. By the mean value theorem, $|T(x) - T(y)| \leq (1/2)|x - y|$. Since $[0, 1]$ is complete, T has a unique fixed point and iteration converges to it.

Exercise 029 – Contraction without completeness

Exercise: ★ ★ ★★

On $X = (0, 1)$ define $T(x) = x/2$. It is a contraction. Does it have a fixed point in X ? What hypothesis is missing?

Solution 029 – Contraction without completeness

Solution:

For $x, y \in (0, 1)$, $|T(x) - T(y)| = \frac{1}{2} |x - y|$, so T is a contraction and maps $(0, 1)$ into itself. A fixed point would satisfy $x = x/2$, hence $x = 0$, but $0 \notin (0, 1)$. The missing hypothesis is completeness: $(0, 1)$ is not complete.

Exercise 030 – Total boundedness in $\mathbb{R}$

Exercise: ★ ★ ★

Show that $[0, 1]$ is totally bounded.

Solution 030 – Total boundedness in $\mathbb{R}$

Solution:

Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ with $1/N < \varepsilon$. The finitely many intervals centered at k/N , $k = 0, \dots, N$, cover $[0, 1]$ with radius ε . Therefore finitely many ε -balls cover $[0, 1]$, so it is totally bounded.

Exercise 031 – Unit ball in infinite dimension

Exercise: ★★★★★

In ℓ^2 , let $e_n = (0, \dots, 0, 1, 0, \dots)$. Show that the closed unit ball is not compact.

Solution 031 – Unit ball in infinite dimension

Solution:

Each e_n has norm 1, so all lie in the closed unit ball. For $m \neq n$, $\|e_n - e_m\|_2 = \sqrt{2}$. Hence no subsequence can be Cauchy, and therefore no subsequence can converge in norm. Sequential compactness fails, so the closed unit ball is not compact.

Exercise 032 – Equivalent norms in finite dimension

Exercise: ★ ★ ★ ★

In $\mathbb{R}^2$, prove $\|x\|_{\infty} \leq \|x\|_2 \leq \sqrt{2} \|x\|_{\infty}$.

Solution 032 – Equivalent norms in finite dimension

Solution:

Let $x = (x_1, x_2)$. Since $\max\{|x_1|, |x_2|\}^2 \leq x_1^2 + x_2^2$, we get $\|x\|_\infty \leq \|x\|_2$. Also $x_1^2 + x_2^2 \leq 2 \max\{|x_1|, |x_2|\}^2$, so $\|x\|_2 \leq \sqrt{2} \|x\|_\infty$.

Exercise 033 – Closed and bounded not compact

Exercise: ★★★★★

In $C[0, 1]$ with the sup norm, consider $f_n(t) = t^n$. Explain why $\{f_n : n \in \mathbb{N}\}$ is bounded but has no uniformly convergent subsequence.

Solution 033 – Closed and bounded not compact

Solution:

Each $\|f_n\|_\infty = 1$, so the set is bounded. Pointwise, $f_n(t) \rightarrow 0$ for $t \in [0, 1)$ and $f_n(1) = 1$. Any uniformly convergent subsequence would converge pointwise to a continuous limit, but the pointwise limit is discontinuous. Hence no uniformly convergent subsequence exists.

Exercise 034 – Incomplete integral norm

Exercise: ★★★★★

On $C[0, 1]$, define $\|f\|_1 = \int_0^1 |f(t)| dt$. Explain why this normed space is not complete.

Solution 034 – Incomplete integral norm

Solution:

Take continuous functions f_n approximating the step function $f = \mathbf{1}_{[1/2,1]}$, with a linear transition of width $1/n$ around $1/2$. Then (f_n) is Cauchy in the integral norm because $f_n \rightarrow f$ in L^1 . But f is not continuous, so the limit is not in $C[0,1]$. Thus $C[0,1]$ is not complete under $\|\cdot\|_1$.

Exercise 035 – Uniform limit theorem

Exercise: ★★ ★

Let $f_n \in C[0, 1]$ and suppose $f_n \rightarrow f$ uniformly. Prove that f is continuous.

Solution 035 – Uniform limit theorem

Solution:

Fix x_0 and $\varepsilon > 0$. Choose N so that $\|f_N - f\|_\infty < \varepsilon/3$. Since f_N is continuous, choose $\delta > 0$ such that $|x - x_0| < \delta$ implies $|f_N(x) - f_N(x_0)| < \varepsilon/3$. Then use the triangle inequality to get $|f(x) - f(x_0)| < \varepsilon$.

Module 1 Exercise Pack

Normed spaces, sequence spaces and inequalities

Exercises 36–60

Exercise 036 – Norm axioms in $\mathbb{R}^2$

Exercise: ★★

Show that $\|x\|_1 = |x_1| + |x_2|$ is a norm on $\mathbb{R}^2$.

Solution 036 – Norm axioms in $\mathbb{R}^2$

Solution:

Non-negativity is clear. If $\|x\|_1 = 0$, then both absolute values are zero, so $x = 0$. Homogeneity: $\|\lambda x\|_1 = |\lambda| \|x\|_1$. Triangle inequality follows from $|x_i + y_i| \leq |x_i| + |y_i|$ for $i = 1, 2$ and summing.

Exercise 037 – Not a norm

Exercise: ★★

On $\mathbb{R}^2$, define $p(x) = x_1^2 + x_2^2$. Which norm axiom fails?

Solution 037 – Not a norm

Solution:

Positive definiteness holds and $p(x) \geq 0$. But homogeneity fails: $p(\lambda x) = \lambda^2 p(x)$, while a norm would require $p(\lambda x) = |\lambda| p(x)$. For $\lambda = 2$ and $x \neq 0$, $4p(x) \neq 2p(x)$.

Exercise 038 – A quasi-norm

Exercise: ★ ★ ★

On $\mathbb{R}^2$, define $q(x) = \sqrt{|x_1|} + \sqrt{|x_2|}$. Is it a norm?

Solution 038 – A quasi-norm

Solution:

It is nonnegative and vanishes only at zero. But homogeneity fails: $q(\lambda x) = \sqrt{|\lambda|}q(x)$, not $|\lambda|q(x)$. Therefore q is not a norm.

Exercise 039 – Unit balls

Exercise: ★★

Draw or describe the unit balls in $\mathbb{R}^2$ for $\|\cdot\|_1$, $\|\cdot\|_2$ and $\|\cdot\|_\infty$.

Solution 039 – Unit balls

Solution:

For $\|\cdot\|_1$, the unit ball is $|x_1| + |x_2| \leq 1$, a diamond. For $\|\cdot\|_2$, it is $x_1^2 + x_2^2 \leq 1$, a disk. For $\|\cdot\|_\infty$, it is $\max\{|x_1|, |x_2|\} \leq 1$, a square.

Exercise 040 – Compute three norms

Exercise: ★

For $x = (2, -1, 2) \in \mathbb{R}^3$, compute $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$.

Solution 040 – Compute three norms

Solution:

$$\|x\|_1 = 2 + 1 + 2 = 5. \quad \|x\|_2 = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{9} = 3. \quad \|x\|_\infty = \max\{2, 1, 2\} = 2.$$

Exercise 041 – Finite-dimensional convergence

Exercise: ★ ★ ★

Let $x_n = (1/n, 1/n^2)$. Show that convergence to 0 holds under $\|\cdot\|_1$, $\|\cdot\|_2$ and $\|\cdot\|_\infty$.

Solution 041 – Finite-dimensional convergence

Solution:

Compute $\|x_n\|_\infty = 1/n \rightarrow 0$. Since $\|x_n\|_2 \leq \|x_n\|_1 = 1/n + 1/n^2 \rightarrow 0$, convergence also holds in ℓ^1 and ℓ^2 norms. Alternatively, in finite dimension all norms are equivalent.

Exercise 042 – Cauchy-Schwarz numerical

Exercise: ★★

For $x = (1, 2, 2)$ and $y = (2, 0, 1)$, verify Cauchy-Schwarz numerically.

Solution 042 – Cauchy-Schwarz numerical

Solution:

Compute $\langle x, y \rangle = 1 \cdot 2 + 2 \cdot 0 + 2 \cdot 1 = 4$. Also $\|x\|_2 = 3$ and $\|y\|_2 = \sqrt{5}$. Thus $|\langle x, y \rangle| = 4 \leq 3\sqrt{5} \approx 6.708$.

Exercise 043 – Equality in Cauchy-Schwarz

Exercise: ★ ★ ★

For which $a \in \mathbb{R}$ do $x = (1, a)$ and $y = (2, 4)$ give equality in Cauchy-Schwarz?

Solution 043 – Equality in Cauchy-Schwarz

Solution:

Equality holds iff the two vectors are linearly dependent. Thus $(1, a) = \lambda(2, 4)$. From the first coordinate, $\lambda = 1/2$. Then $a = \lambda 4 = 2$. Hence equality holds only for $a = 2$.

Exercise 044 – Minkowski in $\mathbb{R}^2$

Exercise: ★★

Verify Minkowski's inequality for $x = (1, 2)$ and $y = (3, -1)$ using the Euclidean norm.

Solution 044 – Minkowski in $\mathbb{R}^2$

Solution:

$x + y = (4, 1)$, so $\|x + y\|_2 = \sqrt{17}$. Also $\|x\|_2 = \sqrt{5}$ and $\|y\|_2 = \sqrt{10}$. Since $\sqrt{17} \leq \sqrt{5} + \sqrt{10}$, Minkowski holds in this example.

Exercise 045 – Sequence in ℓ^2 not ℓ^1

Exercise: ★ ★ ★

Show that $x = (1/n)_{n \geq 1}$ belongs to ℓ^2 but not to ℓ^1 .

Solution 045 – Sequence in ℓ^2 not ℓ^1

Solution:

For ℓ^2 , $\sum_{n=1}^{\infty} (1/n)^2$ converges. For ℓ^1 , $\sum_{n=1}^{\infty} 1/n$ diverges. Therefore $x \in \ell^2$ but $x \notin \ell^1$.

Exercise 046 – Finite support approximation

Exercise: ★ ★ ★

Let $x \in \ell^2$ and define $x^{(N)} = (x_1, \dots, x_N, 0, 0, \dots)$. Prove $x^{(N)} \rightarrow x$ in ℓ^2 .

Solution 046 – Finite support approximation

Solution:

Compute $\|x - x^{(N)}\|_2^2 = \sum_{n>N} |x_n|^2$. Since $x \in \ell^2$, the tail of the convergent series tends to zero. Hence $\|x - x^{(N)}\|_2 \rightarrow 0$.

Exercise 047 – The space c_0

Exercise: ★★☆☆

Let c_0 be the space of sequences converging to 0, with the sup norm. Show that c_0 is closed in ℓ^∞ .

Solution 047 – The space c_0

Solution:

Let $x^{(k)} \in c_0$ and $x^{(k)} \rightarrow x$ in sup norm. Given $\varepsilon > 0$, choose k such that $\|x - x^{(k)}\|_\infty < \varepsilon/2$. Since $x^{(k)} \in c_0$, choose N such that $n \geq N$ implies $|x_n^{(k)}| < \varepsilon/2$. Then $|x_n| \leq \varepsilon$ for $n \geq N$. Thus $x \in c_0$.

Exercise 048 – Completeness of ℓ^∞

Exercise: ★ ★ ★ ★

Explain why ℓ^∞ with the sup norm is complete.

Solution 048 – Completeness of ℓ^∞

Solution:

Let $(x^{(k)})$ be Cauchy in sup norm. For each coordinate n , $(x_n^{(k)})_k$ is Cauchy in $\mathbb{R}$, so it converges to some x_n . Uniform Cauchy control gives a uniform bound, so $x \in \ell^\infty$. Then pass to the coordinatewise limit in the sup estimate to get $\|x^{(k)} - x\|_\infty \rightarrow 0$.

Exercise 049 – Completeness of ℓ^1

Exercise: ★★★★★

Sketch the proof that ℓ^1 is complete.

Solution 049 – Completeness of ℓ^1

Solution:

Let $(x^{(k)})$ be Cauchy in ℓ^1 . Each coordinate is Cauchy, hence converges to x_n . Fatou's lemma or monotone arguments show $\sum |x_n| < \infty$. Finally, for fixed k and large m , $\sum |x_n^{(k)} - x_n^{(m)}| < \varepsilon$; let $m \rightarrow \infty$ to get $\|x^{(k)} - x\|_1 \leq \varepsilon$.

Exercise 050 – Shift operator

Exercise: ★ ★ ★

On ℓ^2 , define $S(x_1, x_2, \dots) = (0, x_1, x_2, \dots)$. Show that S is bounded and compute $\|S\|$.

Solution 050 – Shift operator

Solution:

$\|Sx\|_2^2 = 0^2 + \sum_{n=1}^{\infty} |x_n|^2 = \|x\|_2^2$. Thus $\|Sx\|_2 = \|x\|_2$ for every x . Hence S is bounded and $\|S\| = 1$.

Exercise 051 – Diagonal operator

Exercise: ★ ★ ★★

On ℓ^2 , let $D_a x = (a_n x_n)$, where (a_n) is bounded. Show D_a is bounded and find an upper bound for its norm.

Solution 051 – Diagonal operator

Solution:

Let $M = \sup_n |a_n| < \infty$. Then $\|D_a x\|_2^2 = \sum |a_n x_n|^2 \leq M^2 \sum |x_n|^2 = M^2 \|x\|_2^2$. Thus $\|D_a x\|_2 \leq M \|x\|_2$, so D_a is bounded and $\|D_a\| \leq M$. In fact equality holds by testing near coordinates where $|a_n|$ is close to M .

Exercise 052 – A bounded functional on ℓ^2

Exercise: ★ ★ ★

Define $L(x) = \sum_{n=1}^{\infty} x_n/n$ on ℓ^2 . Prove L is bounded.

Solution 052 – A bounded functional on ℓ^2

Solution:

Use Cauchy-Schwarz with $a_n = 1/n$: $|L(x)| \leq \left(\sum |x_n|^2\right)^{1/2} \left(\sum 1/n^2\right)^{1/2}$. Since $\sum 1/n^2 < \infty$, $|L(x)| \leq C \|x\|_2$. Hence L is bounded.

Exercise 053 – Functional on ℓ^1

Exercise: ★ ★ ★

On ℓ^1 , define $L(x) = \sum_{n=1}^{\infty} x_n$. Show that L is bounded and compute $\|L\|$.

Solution 053 – Functional on ℓ^1

Solution:

For $x \in \ell^1$, $|L(x)| \leq \sum |x_n| = \|x\|_1$, so $\|L\| \leq 1$. Taking $x = e_1 = (1, 0, \dots)$ gives $L(e_1) = 1$ and $\|e_1\|_1 = 1$, hence $\|L\| = 1$.

Exercise 054 – Unbounded linear functional

Exercise: ★★★★★

On the vector space of polynomials with the sup norm on $[0, 1]$, define $L(p) = p'(0)$. Show L is not bounded.

Solution 054 – Unbounded linear functional

Solution:

Take $p_n(t) = t(1 - t)^n$. Then $p'_n(0) = 1$, so $L(p_n) = 1$ for every n . Moreover $\|p_n\|_\infty \rightarrow 0$ because the maximum is of order $1/n$. If L were bounded, $1 \leq C \|p_n\|_\infty \rightarrow 0$, impossible. Hence L is not bounded.

Exercise 055 – Continuity of linear maps

Exercise: ★★

Let $T : X \rightarrow Y$ be linear between normed spaces. Show that if T is bounded, then it is continuous at 0.

Solution 055 – Continuity of linear maps

Solution:

Bounded means $\|Tx\| \leq C \|x\|$ for all x . Given $\varepsilon > 0$, choose $\delta = \varepsilon/C$ if $C > 0$. If $\|x\| < \delta$, then $\|Tx\| < \varepsilon$. If $C = 0$, the result is immediate. Thus T is continuous at 0.

Exercise 056 – Continuity everywhere

Exercise: ★ ★ ★

Show that a linear map continuous at 0 is continuous everywhere.

Solution 056 – Continuity everywhere

Solution:

Let $x_n \rightarrow x$. Then $x_n - x \rightarrow 0$. By continuity at 0, $T(x_n - x) \rightarrow 0$. By linearity, $T(x_n) - T(x) = T(x_n - x) \rightarrow 0$. Hence T is continuous at x . Since x was arbitrary, T is continuous everywhere.

Exercise 057 – Sup norm on functions

Exercise: ★

For $f(t) = t^2$ on $[0, 1]$, compute $\|f\|_\infty$.

Solution 057 – Sup norm on functions

Solution:

Since t^2 is increasing on $[0, 1]$, its maximum absolute value is attained at $t = 1$. Thus $\|f\|_{\infty} = \sup_{t \in [0, 1]} t^2 = 1$.

Exercise 058 – Distance between functions

Exercise: ★★

In $C[0, 1]$ with the sup norm, compute $d_{\infty}(f, g)$ for $f(t) = t$ and $g(t) = t^2$.

Solution 058 – Distance between functions

Solution:

$d_{\infty}(f, g) = \sup_{t \in [0, 1]} |t - t^2| = \sup t(1 - t)$. The maximum occurs at $t = 1/2$, with value $1/4$. Hence $d_{\infty}(f, g) = 1/4$.

Exercise 059 – Pointwise not uniform

Exercise: ★ ★ ★

Show that $f_n(t) = t^n$ on $[0, 1]$ converges pointwise but not uniformly.

Solution 059 – Pointwise not uniform

Solution:

For $t \in [0, 1)$, $t^n \rightarrow 0$, while $f_n(1) = 1$. The pointwise limit is $f(t) = 0$ for $t < 1$ and $f(1) = 1$, which is discontinuous. Since a uniform limit of continuous functions is continuous, convergence cannot be uniform.

Exercise 060 – Uniform convergence

Exercise: ★★

Show that $f_n(t) = t/n$ converges uniformly to 0 on $[0, 1]$.

Solution 060 – Uniform convergence

Solution:

Compute $\|f_n - 0\|_\infty = \sup_{t \in [0,1]} t/n = 1/n$. Since $1/n \rightarrow 0$, the convergence is uniform.

Module 1 Exercise Pack

Function spaces and linear operators

Exercises 61–80

Exercise 061 – Integral norm is a norm

Exercise: ★ ★ ★

On $C[0, 1]$, define $\|f\|_1 = \int_0^1 |f(t)| dt$. Prove that this is a norm.

Solution 061 – Integral norm is a norm

Solution:

Non-negativity is clear. If $\|f\|_1 = 0$, then $|f(t)|$ is continuous and has integral zero; if it were positive somewhere, continuity would give positive integral. Thus $f = 0$. Homogeneity is immediate. Triangle inequality follows from $|f + g| \leq |f| + |g|$ and integration.

Exercise 062 – Evaluation under sup norm

Exercise: ★★

Let $L(f) = f(1/2)$ on $C[0, 1]$ with the sup norm. Show L is bounded and compute $\|L\|$.

Solution 062 – Evaluation under sup norm

Solution:

$|L(f)| = |f(1/2)| \leq \|f\|_\infty$, so $\|L\| \leq 1$. Taking the constant function $f(t) = 1$ gives $|L(f)| = 1$ and $\|f\|_\infty = 1$. Hence $\|L\| = 1$.

Exercise 063 – Evaluation under integral norm

Exercise: ★★★★★

On $C[0, 1]$ with $\|f\|_1 = \int_0^1 |f(t)| dt$, show $L(f) = f(0)$ is not bounded.

Solution 063 – Evaluation under integral norm

Solution:

Construct triangular functions f_n with $f_n(0) = 1$, decreasing linearly to 0 on $[0, 1/n]$, and equal to 0 afterwards. Then $\|f_n\|_1 = 1/(2n) \rightarrow 0$, but $L(f_n) = 1$. Boundedness would imply $1 \leq C/(2n)$ for all n , impossible.

Exercise 064 – Integral operator

Exercise: ★ ★ ★

Define $Tf(x) = \int_0^x f(t)dt$ on $C[0, 1]$ with the sup norm. Prove T is bounded.

Solution 064 – Integral operator

Solution:

For each $x \in [0, 1]$, $|Tf(x)| \leq \int_0^x |f(t)| dt \leq x \|f\|_\infty \leq \|f\|_\infty$. Taking the supremum over x gives $\|Tf\|_\infty \leq \|f\|_\infty$. Hence T is bounded with $\|T\| \leq 1$.

Exercise 065 – Norm of integral operator

Exercise: ★★☆☆

For the operator $Tf(x) = \int_0^x f(t)dt$, show that $\|T\| = 1$ on $C[0, 1]$ with sup norm.

Solution 065 – Norm of integral operator

Solution:

The previous exercise gives $\|T\| \leq 1$. Take $f(t) = 1$. Then $\|f\|_\infty = 1$ and $Tf(x) = x$, so $\|Tf\|_\infty = 1$. Therefore $\|T\| \geq 1$. Hence $\|T\| = 1$.

Exercise 066 – Multiplication operator

Exercise: ★ ★ ★

Let $m(t) = t$ and define $Mf(t) = m(t)f(t)$ on $C[0, 1]$. Compute $\|M\|$.

Solution 066 – Multiplication operator

Solution:

For all f , $\|Mf\|_{\infty} = \sup |tf(t)| \leq \sup |f(t)| = \|f\|_{\infty}$, so $\|M\| \leq 1$. Taking $f(t) = 1$ gives $Mf(t) = t$ and $\|Mf\|_{\infty} = 1$. Thus $\|M\| = 1$.

Exercise 067 – Derivative unbounded

Exercise: ★★★★★

Let D be the derivative operator from polynomials with sup norm on $[0, 1]$ to $C[0, 1]$. Show D is not bounded.

Solution 067 – Derivative unbounded

Solution:

Take $p_n(t) = t^n$. Then $\|p_n\|_\infty = 1$. But $Dp_n(t) = nt^{n-1}$, so $\|Dp_n\|_\infty = n$. If D were bounded, we would have $n \leq C$ for all n , impossible.

Exercise 068 – Derivative bounded with stronger norm

Exercise: ★ ★ ★

On $C^1[0, 1]$, define $\|f\|_{C^1} = \|f\|_{\infty} + \|f'\|_{\infty}$. Show $D(f) = f'$ is bounded.

Solution 068 – Derivative bounded with stronger norm

Solution:

By definition, $\|Df\|_{\infty} = \|f'\|_{\infty} \leq \|f\|_{\infty} + \|f'\|_{\infty} = \|f\|_{C^1}$. Hence D is bounded and $\|D\| \leq 1$ as a map from $(C^1, \|\cdot\|_{C^1})$ to $(C, \|\cdot\|_{\infty})$.

Exercise 069 – A closed subspace

Exercise: ★★

Show that $M = \{f \in C[0, 1] : f(0) = 0\}$ is a closed subspace of $C[0, 1]$ with the sup norm.

Solution 069 – A closed subspace

Solution:

It is a subspace because evaluation at 0 is linear. Let $f_n \in M$ and $f_n \rightarrow f$ uniformly. Then $f_n(0) \rightarrow f(0)$, but $f_n(0) = 0$ for all n , so $f(0) = 0$. Hence $f \in M$. Therefore M is closed.

Exercise 070 – Integral-zero subspace

Exercise: ★★☆☆

Show that $M = \{f \in C[0, 1] : \int_0^1 f(t)dt = 0\}$ is closed in the sup norm.

Solution 070 – Integral-zero subspace

Solution:

Define $L(f) = \int_0^1 f(t)dt$. Then L is linear and $|L(f)| \leq \int_0^1 |f(t)| dt \leq \|f\|_\infty$, so L is continuous. The set M is $\text{Ker } L = L^{-1}(\{0\})$. Since $\{0\}$ is closed and L is continuous, M is closed.

Exercise 071 – Rank-one operator

Exercise: ★ ★ ★

Let $Tf = f(0) \cdot 1$, where 1 denotes the constant function. Compute $\|T\|$ on $C[0, 1]$ with sup norm.

Solution 071 – Rank-one operator

Solution:

$\|Tf\|_\infty = |f(0)| \|1\|_\infty = |f(0)| \leq \|f\|_\infty$, so $\|T\| \leq 1$. Taking $f = 1$ gives $\|Tf\|_\infty = 1 = \|f\|_\infty$. Hence $\|T\| = 1$.

Exercise 072 – Matrix norm induced by ℓ^∞

Exercise: ★ ★ ★★

For $A = \begin{pmatrix} 1 & -2 \\ 3 & 1 \end{pmatrix}$, compute the operator norm induced by $\|\cdot\|_\infty$.

Solution 072 – Matrix norm induced by ℓ^∞

Solution:

For the ℓ^∞ induced norm, $\|A\|_\infty$ is the maximum absolute row sum. Row sums are $|1| + |-2| = 3$ and $|3| + |1| = 4$. Hence $\|A\|_\infty = 4$.

Exercise 073 – Matrix norm induced by ℓ^1

Exercise: ★ ★ ★★

For $A = \begin{pmatrix} 1 & -2 \\ 3 & 1 \end{pmatrix}$, compute the operator norm induced by $\|\cdot\|_1$.

Solution 073 – Matrix norm induced by ℓ^1

Solution:

For the ℓ^1 induced norm, $\|A\|_1$ is the maximum absolute column sum. Column sums are $|1| + |3| = 4$ and $|-2| + |1| = 3$. Hence $\|A\|_1 = 4$.

Exercise 074 – Kernel of bounded operator

Exercise: ★★

Let $T : X \rightarrow Y$ be bounded and linear. Show that $\text{Ker } T$ is closed.

Solution 074 – Kernel of bounded operator

Solution:

Let $x_n \in \text{Ker } T$ and $x_n \rightarrow x$ in X . Since T is bounded, it is continuous. Therefore $Tx_n \rightarrow Tx$. But $Tx_n = 0$ for all n , so $Tx = 0$. Hence $x \in \text{Ker } T$, and $\text{Ker } T$ is closed.

Exercise 075 – Bounded iff continuous

Exercise: ★★☆☆

Prove that a linear map between normed spaces is continuous iff it is bounded.

Solution 075 – Bounded iff continuous

Solution:

Bounded implies continuous by $\|Tx\| \leq C \|x\|$. Conversely, if T is continuous at 0, choose $\delta > 0$ such that $\|x\| < \delta$ implies $\|Tx\| < 1$. For $x \neq 0$, set $u = \delta x / (2 \|x\|)$. Then $\|u\| < \delta$, so $\|Tu\| < 1$. By linearity, $\|Tx\| < 2 \|x\| / \delta$.

Exercise 076 – Inner product axioms

Exercise: ★★

Verify that $\langle x, y \rangle = x_1 y_1 + 2x_2 y_2$ is an inner product on $\mathbb{R}^2$.

Solution 076 – Inner product axioms

Solution:

It is symmetric and bilinear. Positivity: $\langle x, x \rangle = x_1^2 + 2x_2^2 \geq 0$. If $\langle x, x \rangle = 0$, then both $x_1 = 0$ and $x_2 = 0$. Hence it is positive definite. Therefore it is an inner product.

Exercise 077 – Not an inner product

Exercise: ★★

Show that $\langle x, y \rangle = x_1 y_1 - x_2 y_2$ is not an inner product on $\mathbb{R}^2$.

Solution 077 – Not an inner product

Solution:

The problem is positivity. Take $x = (0, 1)$. Then $\langle x, x \rangle = 0^2 - 1^2 = -1 < 0$. An inner product must satisfy $\langle x, x \rangle \geq 0$ for all x . Therefore this formula is not an inner product.

Exercise 078 – Weighted norm

Exercise: ★ ★ ★

For $\langle x, y \rangle = x_1 y_1 + 2x_2 y_2$, compute the induced norm of $x = (2, 3)$.

Solution 078 – Weighted norm

Solution:

The induced norm is $\|x\| = \sqrt{\langle x, x \rangle}$. Here $\langle x, x \rangle = 2^2 + 2 \cdot 3^2 = 4 + 18 = 22$. Therefore $\|x\| = \sqrt{22}$.

Exercise 079 – Parallelogram identity

Exercise: ★ ★ ★

Verify the parallelogram identity for $x = (1, 0)$ and $y = (0, 2)$ in Euclidean space.

Solution 079 – Parallelogram identity

Solution:

$x + y = (1, 2)$ and $x - y = (1, -2)$. Thus $\|x + y\|^2 = 5$ and $\|x - y\|^2 = 5$, so the left side is 10. The right side is $2\|x\|^2 + 2\|y\|^2 = 2 \cdot 1 + 2 \cdot 4 = 10$.

Exercise 080 – Pythagoras

Exercise: ★★

Let $x = (1, 2)$ and $y = (2, -1)$. Check that $x \perp y$ and verify Pythagoras.

Solution 080 – Pythagoras

Solution:

$\langle x, y \rangle = 1 \cdot 2 + 2 \cdot (-1) = 0$, so $x \perp y$. $x + y = (3, 1)$, so $\|x + y\|^2 = 10$. Also $\|x\|^2 = 5$ and $\|y\|^2 = 5$. Hence $\|x + y\|^2 = \|x\|^2 + \|y\|^2$.

Module 1 Exercise Pack

Inner product and Hilbert spaces

Exercises 81–100

Exercise 081 – Projection onto a line

Exercise: ★ ★ ★

Project $x = (3, 1)$ onto the line spanned by $u = (1, 1)$ in $\mathbb{R}^2$.

Solution 081 – Projection onto a line

Solution:

The projection is $P_M x = \frac{\langle x, u \rangle}{\langle u, u \rangle} u$. Compute $\langle x, u \rangle = 4$ and $\langle u, u \rangle = 2$. Hence $P_M x = 2(1, 1) = (2, 2)$. The error is $x - P_M x = (1, -1)$, orthogonal to u .

Exercise 082 – Distance to a line

Exercise: ★ ★ ★

Find the distance from $x = (3, 1)$ to the line $M = \text{span}\{(1, 1)\}$.

Solution 082 – Distance to a line

Solution:

From the previous projection, $P_M x = (2, 2)$. The distance is the norm of the orthogonal error: $\text{dist}(x, M) = \|x - P_M x\| = \|(1, -1)\| = \sqrt{2}$.

Exercise 083 – Projection onto a plane

Exercise: ★ ★ ★

In $\mathbb{R}^3$, project $x = (1, 2, 3)$ onto the plane $M = \{(a, b, 0) : a, b \in \mathbb{R}\}$.

Solution 083 – Projection onto a plane

Solution:

The plane consists of vectors with third coordinate zero. The closest point keeps the first two coordinates and sets the third to zero: $P_M x = (1, 2, 0)$. The error is $(0, 0, 3)$, orthogonal to every vector $(a, b, 0)$ in the plane.

Exercise 084 – Gram-Schmidt

Exercise: ★ ★ ★★

Apply Gram-Schmidt to $v_1 = (1, 1)$ and $v_2 = (1, 0)$ in $\mathbb{R}^2$.

Solution 084 – Gram-Schmidt

Solution:

First $e_1 = v_1 / \|v_1\| = (1, 1) / \sqrt{2}$. Remove the component of v_2 along e_1 : $w_2 = v_2 - \langle v_2, e_1 \rangle e_1 = (1, 0) - (1/\sqrt{2})(1, 1)/\sqrt{2} = (1/2, -1/2)$. Normalize: $e_2 = w_2 / \|w_2\| = (1, -1) / \sqrt{2}$.

Exercise 085 – Orthogonal complement

Exercise: ★ ★ ★

Find $M^\perp$ for $M = \text{span}\{(1, 2, 0)\}$ in $\mathbb{R}^3$.

Solution 085 – Orthogonal complement

Solution:

A vector $x = (a, b, c)$ belongs to $M^\perp$ iff $\langle x, (1, 2, 0) \rangle = a + 2b = 0$. Hence $a = -2b$, and c is free. Therefore $M^\perp = \{(-2b, b, c) : b, c \in \mathbb{R}\}$.

Exercise 086 – Bessel inequality finite

Exercise: ★ ★ ★

Let $e_1 = (1, 0, 0)$ and $e_2 = (0, 1, 0)$ in $\mathbb{R}^3$. Verify Bessel's inequality for $x = (1, 2, 3)$.

Solution 086 – Bessel inequality finite

Solution:

The left side is $|\langle x, e_1 \rangle|^2 + |\langle x, e_2 \rangle|^2 = 1^2 + 2^2 = 5$. The norm squared of x is $1^2 + 2^2 + 3^2 = 14$. Bessel says $5 \leq 14$, which holds.

Exercise 087 – Parseval in an orthonormal basis

Exercise: ★★

For the standard basis of $\mathbb{R}^3$, verify Parseval for $x = (1, -2, 2)$.

Solution 087 – Parseval in an orthonormal basis

Solution:

The coefficients are $1, -2, 2$. Parseval gives $\|x\|^2 = 1^2 + (-2)^2 + 2^2 = 9$. Directly, $\|x\|^2 = 1 + 4 + 4 = 9$.

Exercise 088 – Projection theorem intuition

Exercise: ★ ★ ★

Let M be a closed subspace of a Hilbert space. State the projection theorem and identify the orthogonality condition.

Solution 088 – Projection theorem intuition

Solution:

For every x , there exists a unique $m = P_M x \in M$ such that $\|x - m\| = \text{dist}(x, M)$. The error $x - P_M x$ is orthogonal to the subspace: $x - P_M x \in M^\perp$. Thus $x = P_M x + (x - P_M x)$ is an orthogonal decomposition.

Exercise 089 – Closed convex projection in $\mathbb{R}^2$

Exercise: ★★☆☆

Find the closest point in the closed half-line $C = \{(t, 0) : t \geq 0\}$ to $x = (-1, 2)$.

Solution 089 – Closed convex projection in $\mathbb{R}^2$

Solution:

A point in C has form $(t, 0)$ with $t \geq 0$. The squared distance is $(t + 1)^2 + 4$. This is minimized at $t = -1$ without the constraint, but the constraint requires $t \geq 0$. The nearest feasible value is $t = 0$. Thus the closest point is $(0, 0)$.

Exercise 090 – Nonclosed subspace and projection

Exercise: ★★★★★

Let c_{00} be the finite-support sequences in ℓ^2 . Explain why projection onto c_{00} may fail.

Solution 090 – Nonclosed subspace and projection

Solution:

The subspace c_{00} is dense in ℓ^2 but not closed. Take $x = (1/n)_{n \geq 1} \in \ell^2$. It is not in c_{00} . Since finite truncations of x lie in c_{00} and converge to x , the distance from x to c_{00} is 0. But no vector in c_{00} equals x , so the distance is not attained. Projection fails.

Exercise 091 – Incomplete inner product space

Exercise: ★★★★★

Explain why $C[0, 1]$ with inner product $\langle f, g \rangle = \int_0^1 f(t)g(t)dt$ is not a Hilbert space.

Solution 091 – Incomplete inner product space

Solution:

The induced norm is the L^2 norm. Continuous functions can be Cauchy in L^2 and converge to a discontinuous square-integrable function, for example approximations of a step function. The limit is not in $C[0, 1]$. Thus the space is not complete under this norm, so it is not Hilbert.

Exercise 092 – ℓ^2 is Hilbert

Exercise: ★★☆☆

Why is ℓ^2 a Hilbert space?

Solution 092 – ℓ^2 is Hilbert

Solution:

Define $\langle x, y \rangle = \sum_{n=1}^{\infty} x_n y_n$. Cauchy-Schwarz guarantees the series is absolutely controlled and defines an inner product. The induced norm is the usual ℓ^2 norm. Since ℓ^2 is complete under this norm, it is a Hilbert space.

Exercise 093 – Least squares as projection

Exercise: ★ ★ ★★

Fit a constant c to the data $(1, 2, 4)$ by minimizing $(1 - c)^2 + (2 - c)^2 + (4 - c)^2$.

Solution 093 – Least squares as projection

Solution:

Differentiate: $F(c) = (1 - c)^2 + (2 - c)^2 + (4 - c)^2$. Then $F'(c) = 2(c - 1) + 2(c - 2) + 2(c - 4) = 6c - 14$. Set $F'(c) = 0$ to get $c = 14/6 = 7/3$. This is the projection of the vector $(1, 2, 4)$ onto the subspace of constant vectors.

Exercise 094 – Normal equations

Exercise: ★★★★★

Find the least-squares line $a + bt$ through points $(0, 1), (1, 2), (2, 2)$.

Solution 094 – Normal equations

Solution:

Minimize $S(a, b) = (1-a)^2 + (2-a-b)^2 + (2-a-2b)^2$. Normal equations: $\sum (y_i - a - bt_i) = 0$ and $\sum t_i(y_i - a - bt_i) = 0$. Thus $5 - 3a - 3b = 0$ and $6 - 3a - 5b = 0$. Solving gives $b = 1/2$, $a = 7/6$.

Exercise 095 – Exam synthesis: norm and projection

Exercise: ★★★★★

Let $M = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$. Show M is closed and find the projection of $u = (1, 2, 3)$ onto M .

Solution 095 – Exam synthesis: norm and projection

Solution:

M is the kernel of the continuous linear map $L(x, y, z) = x + y + z$, hence closed. The orthogonal complement is $\text{span}\{n\}$ with $n = (1, 1, 1)$. Projection onto M is $u - \frac{\langle u, n \rangle}{\langle n, n \rangle} n$. Since $\langle u, n \rangle = 6$ and $\langle n, n \rangle = 3$, $P_M u = (1, 2, 3) - 2(1, 1, 1) = (-1, 0, 1)$.

Exercise 096 – Parallelogram test for a norm

Exercise: ★★☆☆

On $\mathbb{R}^2$, the norm $\|x\|_1 = |x_1| + |x_2|$ is not induced by an inner product. Verify this using the parallelogram identity.

Solution 096 – Parallelogram test for a norm

Solution:

Take $x = (1, 0)$ and $y = (0, 1)$. Then $\|x + y\|_1^2 = 2^2 = 4$ and $\|x - y\|_1^2 = 2^2 = 4$, so the left side is 8. The right side is $2\|x\|_1^2 + 2\|y\|_1^2 = 2 + 2 = 4$. Since $8 \neq 4$, the parallelogram identity fails, so this norm is not induced by an inner product.

Exercise 097 – Best approximation by a constant

Exercise: ★ ★ ★

In $\mathbb{R}^4$, find the projection of $x = (1, 3, 5, 7)$ onto the subspace of constant vectors (c, c, c, c) .

Solution 097 – Best approximation by a constant

Solution:

The subspace is spanned by $u = (1, 1, 1, 1)$. Projection is $Px = \frac{\langle x, u \rangle}{\langle u, u \rangle} u$. Here $\langle x, u \rangle = 16$ and $\langle u, u \rangle = 4$, hence $c = 4$. The projection is $(4, 4, 4, 4)$.

Exercise 098 – Orthogonal decomposition

Exercise: ★ ★ ★★

Let $M = \text{span}\{(1, 1, 0), (0, 1, 1)\} \subset \mathbb{R}^3$. Find a nonzero vector in $M^\perp$.

Solution 098 – Orthogonal decomposition

Solution:

Let $v = (a, b, c)$. Orthogonality requires $a + b = 0$ and $b + c = 0$. Hence $a = -b$ and $c = -b$. Taking $b = -1$ gives $v = (1, -1, 1)$. Therefore $M^\perp = \text{span}\{(1, -1, 1)\}$.

Exercise 099 – Operator norm from an inner product

Exercise: ★ ★ ★★

Let $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ on Euclidean $\mathbb{R}^2$. Compute $\|A\|$.

Solution 099 – Operator norm from an inner product

Solution:

For $x = (x_1, x_2)$, $\|Ax\|_2^2 = 4x_1^2 + x_2^2 \leq 4(x_1^2 + x_2^2) = 4\|x\|_2^2$, so $\|A\| \leq 2$. Taking $x = (1, 0)$ gives $\|Ax\|_2 = 2 = 2\|x\|_2$. Hence $\|A\| = 2$.

Exercise 100 – Full exam challenge

Exercise: ★★★★★

Let $T : C[0, 1] \rightarrow C[0, 1]$ be $Tf(x) = xf(x)$. Show that T is bounded, compute its norm, and determine whether $\text{Ker } T$ is trivial.

Solution 100 – Full exam challenge

Solution:

Boundedness: $\|Tf\|_{\infty} = \sup_{x \in [0,1]} |xf(x)| \leq \|f\|_{\infty}$, so $\|T\| \leq 1$. With $f \equiv 1$, $\|Tf\|_{\infty} = 1$, hence $\|T\| = 1$. If $Tf = 0$, then $xf(x) = 0$ for every x . For $x > 0$, $f(x) = 0$. By continuity, also $f(0) = 0$. Thus $\text{Ker } T = \{0\}$.

Final checklist for Module 1 exam exercises

Exercise: Before submitting a solution

- ① Did I write the definition used in the proof?
- ② Did I check all assumptions of the theorem?
- ③ If I used compactness, did I know why the set is compact?
- ④ If I used completeness, did I know the space is complete?
- ⑤ If I disproved something, did I provide one explicit counterexample?
- ⑥ If I computed a projection, did I verify orthogonality of the residual?

Warning: Most common error

Do not confuse boundedness with compactness in infinite-dimensional spaces.